

Achint Sharma

Doctoral Researcher | Multisensory Cognition | Audiovisual Speech Processing

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RESEARCH PROFILE

Cognitive science doctoral researcher investigating audiovisual temporal integration in speech, with experience in behavioral experiment design, multisensory perception, EEG methods and experimental research. Experienced working with adult and pediatric populations and translating theoretical questions into controlled experimental paradigms.

RESEARCH INTERESTS

Audiovisual speech perception; multisensory temporal integration; working memory and multisensory binding; temporal recalibration; cognitive neuroscience; EEG and neural oscillations

EDUCATION

Ph.D., Cognitive and Information Sciences

University of California, Merced, Merced, CA

Expected completion: June 2028

Adviser(s): Prof. Kristina Backer; Prof. Heather Bortfeld

Dissertation/research focus: Audiovisual Integration in Speech

Aug 2023–Present

M.S., Cognitive Science

Center of Behavioral and Cognitive Sciences, University of

Allahabad, Allahabad, India

Thesis: Resting-state and Task-based modulations in EEG rhythms related to Gaze in Autism Spectrum Disorder

Adviser: Prof. Bhoomika R. Kar

June 2016–2018

RESEARCH EXPERIENCE

Doctoral Researcher

Cognitive Auditory Neuroscience (CAN) Lab and Bortfeld Lab,

University of California, Merced

Aug 2023–Present

- Investigate audiovisual temporal integration in speech, with emphasis on Speech perception.
- Design and program behavioral paradigms involving audiovisual timing, speech stimuli, and working-memory demands using MATLAB, Psychtoolbox, Python, and Presentation (Neurobehavioral Systems).
- Analyze trial-level behavioral data using R (generalized linear mixed-effects models, multinomial logistic regression, psychometric modeling), *EEGLAB*, *ERPLAB*.
- Develop and validate experimental tasks/stimulus sets; document timing accuracy, response collection, and reproducible analysis workflows.

Research Assistant

Multisensory Processing Lab

University of California, Los Angeles

Mar 2022–May 2023

- Coordinated and conducted behavioral data collection sessions for 150+ participants using Pavlovia and supported standardized administration and data-quality procedures.
- Prepared, organized, and analyzed behavioral datasets using Python from all participants, generating scripts for analysis and contributing to the writing of the paper.
- Contributed to the experimental design of a project on multisensory integration and learning by testing and debugging the audio input required for the piloting of the study using PsychoPy.
- Contributed to a peer-reviewed publication.

Junior Research Fellow / Project Manager

Cognitive and Behavioral Neuroscience Lab,

Indian Institute of Technology Bombay, Mumbai, India

Feb 2019–Dec 2019

- Managed the project “Cognitive Marker of ADHD Subtype: Development of a Diagnostic Tool,” coordinating with the team of 4 research fellows with core responsibilities of reaching out to the collaborators in the field, including special schools and clinics, along with organizing data collection sessions in the field.
- Maintained and analyzed 200+ experimental records using multinomial logistic regression and receiver operating characteristic (ROC) analysis.
- Administered and scored the Conners’ Parent Rating Scale and Gilliam Autism Rating Scale (completed training for administering the scales during MS while internship at Allahabad Child Development Center).
- Trained 3 fellow researchers in standardized data-collection procedures, improving uniformity of administration of experiments in the field.

SELECTED PUBLICATION

Shams, L. T., Förby, A., **Sharma, A.**, & Shams, L. (2023). Big number, big body: Jersey numbers alter body size perception. *PLOS ONE*, 18(9), e0287474. Link: <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0287474>

CONFERENCE PRESENTATIONS

Sharma, A., Shahin, A., Backer, K., & Bortfeld, H. (2025, April). *Effect of context-dependent visual information on the audiovisual time-flow illusion*. Cognitive Neuroscience Society Annual Meeting, Boston.

Sharma, A., Kar, B. R., & Srivastava, C. (2018). *Resting-state and task-based EEG related to gaze in autism spectrum disorder*. 5th Annual Conference of Cognitive Science, IIT Guwahati, India.

TECHNICAL SKILLS

Programming and analysis: Python; MATLAB; R; EEGLAB;

Experimental methods: Behavioral experiment design; audiovisual stimulus design; EEG data collection; time-frequency analysis and ERP analysis using EEGLAB and ERPLAB.

Statistics: Generalized linear mixed-effects models; multinomial logistic regression; ROC analysis; correlation; Bayesian modeling

Research platforms: Psychtoolbox, PsychoPy, Pavlovia, GitHub, Adobe Audition

AWARDS AND FUNDING

- MacKenzie Scott Graduate Student Supplemental Travel Award, University of California, Merced, 2025.
- Merit Scholarship, Center of Behavioral and Cognitive Sciences, 2017 for the duration of 1.5 years.

TEACHING & MENTORING

- Teaching Assistant: TA experience of 6 semesters for research writing, introduction, and Cognitive neuroscience course, including responsibilities of conducting discussion sections and providing assistance to help students make progress, as well as creating and evaluating tests.
- Mentored and trained the current research assistants working on my projects regarding data collection
- Taught for the after-school educational activities with Abhyasika (non-profit) for students from economically disadvantaged backgrounds (2019).
- Participated as an assessment evaluator for a community dementia-screening initiative at the Center of Behavioral and Cognitive Sciences, administering cognitive assessments to older adults (2018).

ADDITIONAL PROFESSIONAL EXPERIENCE

Case Manager (Trainee)

Jan 2020–Jun 2021

Heartland Human Care Services, Chicago, IL

- Coordinated multidisciplinary services for unaccompanied immigrant minors in collaboration with families, clinical psychologists, educators, social workers, and attorneys. Maintained confidential participant records and tracked legal, safety, and mental-health requirements for individuals under the program.

PROFESSIONAL DEVELOPMENT

- Advanced Methods in EEG Data Analysis, EEGLAB Workshop, University of California, San Diego, 2022.
- Attended Developmental Cognitive Neuroscience course taught by Chandan J. Vaidya, Georgetown University, Oct 2018